How We Tested

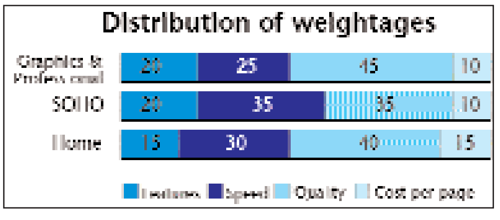
Our test-bed comprised a Pentium 4 3.2 GHz processor plugged onto an Intel 875PBZ motherboard, with 512 MB of Corsair 400 MHz DDR RAM, an MSI Ti4800 display card and a 40 GB 7,200 rpm ATA IV Seagate Barracuda hard disk. We used Windows XP with SP1 as the OS. The system was loaded with all the latest chipset and graphics drivers. We also installed the USB patch for XP, and connected the devices on the USB 2.0 port for optimum performance. Across the tests, we used paper as shown below.

Tests	Paper Used
Normal tests	100 GSM
Photograph printing	NovaJet 130 GSM
Cost per page test	75 GSM

GSM: Grams per square metre
NovaJet: A widely available photo paper brand for inkjets

Having established the three areas—home, SoHo and 'graphics and professional' printing to evaluate the printers, we decided to further define these areas according to the printing qualities and feature sets available.

The graph below shows the individual weightages assigned to the features set, the print speed, quality and cost per page for all the three areas. The ideal home solution would therefore be a printer that gave decent speeds for both text and images, and decent



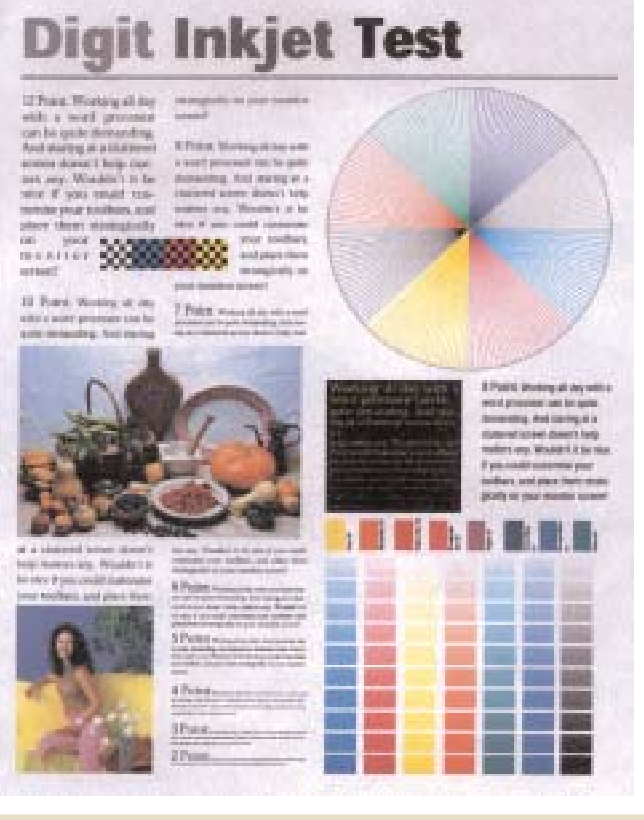


photo printouts, while still being affordable—in the range of Rs 3,000 to Rs 5,000.

In the SoHo category, we assigned equal weightages for speed and quality. This was done with the assumption that in a typical office, a perfect balance of speed and quality is needed, along with features such as two-sided printing, ink status reports through the driver, a larger buffer memory, etc.

Coming to graphics and professional printing solutions, the score split was the same as for office printers, with one difference: In terms of performance, output quality was given more importance than speed.

We tested all the printers on the same test benchmarks. These comprised a text document, a combi-document and an image file. These documents incorporate every small detail that a printer might need to reproduce—the smallest-point text and lines, the specular effect (shine) on metal, etc.

The combi-document had text with varying points: lines drawn inside a circle, seven colour bars of decreasing gradient, and a small picture at one corner. Each of these sub-sections tests a printer's ability to reproduce details. Varying-point text size tests its ability to reproduce the smallest possible text, that is, the spatial resolution of the printer. The circle of lines tests whether the printer can reproduce lines without jarring edges. Similarly, the seven-colour bar test gauges the colour reproduction capabilities of the cartridges.

The image document, too, was made up of several elements, with areas of varying contrast and brightness. This tests a printer's colour reproduction capability on photo paper.

The text and the combi-document were printed at Normal and Best quality settings, whereas for the image file, we used only the Best quality setting.